CERTIFICATION SUCCESS GUIDE

PROGRAMMER FOR JAVA™ 2 PLATFORM

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introduction

GET AHEAD, STAY AHEAD

We developed this Certification Success Guide to help you gain visibility for your most valuable asset — your skills. At Sun, we know that the future starts in the minds of the people that take our technologies to the next level. Our future is you, and we're committed to your success.

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http://suned.sun.com/US/certification/

Java™ technology certification is a valuable, career-building opportunity – and a great way to help you take charge of your career. We're behind you from the start with education, training, and the resources you need to help you reach your certification goals.

Sun Educational Services offers courseware and certification to support you in expanding your career opportunities and creating the systems and applications that fuel the future. Our learning paths provide at-a-glance training and education information, so you know what you need to learn to successfully prepare for exams.

The Benefits of Certification

FOR THE INDIVIDUAL

Becoming certified in Java technology can help you improve your career potential, gain more respect, and increase job security. With certification, you have hard evidence that you're qualified for the tasks that lie ahead, helping you increase your opportunities for professional advancement, such as salary increase, job role modifications, or promotions.

FOR THE CORPORATION

Corporations that employ certified individuals do so to gain a competitive advantage. The skills that are verified during the certification process are the same skills that can help lead to decreased time to market, increased productivity, less system failure, and higher employee satisfaction rates.

Recruiting certified employees and certifying existing employees can lead to a more stable work environment, which in turn can lead to greater success as a whole. When companies demonstrate that they are willing to invest in their employees, those employees can be more productive, more loyal, and are more likely to remain in their jobs.

Certification Requirements Checklist

The Sun Certified Programmer for Java 2 Platform is composed of one multiple-choice exam that covers Java 2 Platform, Standard Edition. You must successfully complete the exam to become certified. The exam details are as follows:

- [ ] **Step 1. The Sun Certified Programmer for Java 2 Platform.**

Available at: Authorized Prometric testing centers
Exam number: 310-025
Prerequisites: None
Exam type: Multiple choice, short answer, and drag and drop
Number of questions: 59
Pass score: 61%
Time limit: 120 minutes
Cost: US$150, or as locally priced

Exam Preparation

To help you prepare, we offer an online practice exam and courseware. The practice exam can help you detect skill gaps and identify the appropriate courses so you can increase knowledge prior to taking the exam. For a free sample practice test, visit http://suned.sun.com/US/certification/java/index.html.

Our courseware offerings provide information to help you pass certification exams and do your job with confidence. For the Sun Certified Programmer for Java 2 Platform, practice exam information and supporting courseware includes:

Practice Certification for the Sun Certified Programmer for Java 2 Platform
Exam number: WGS-PREX-J025
Duration: 30-day subscription
Delivery: Sun™ Web Learning Center

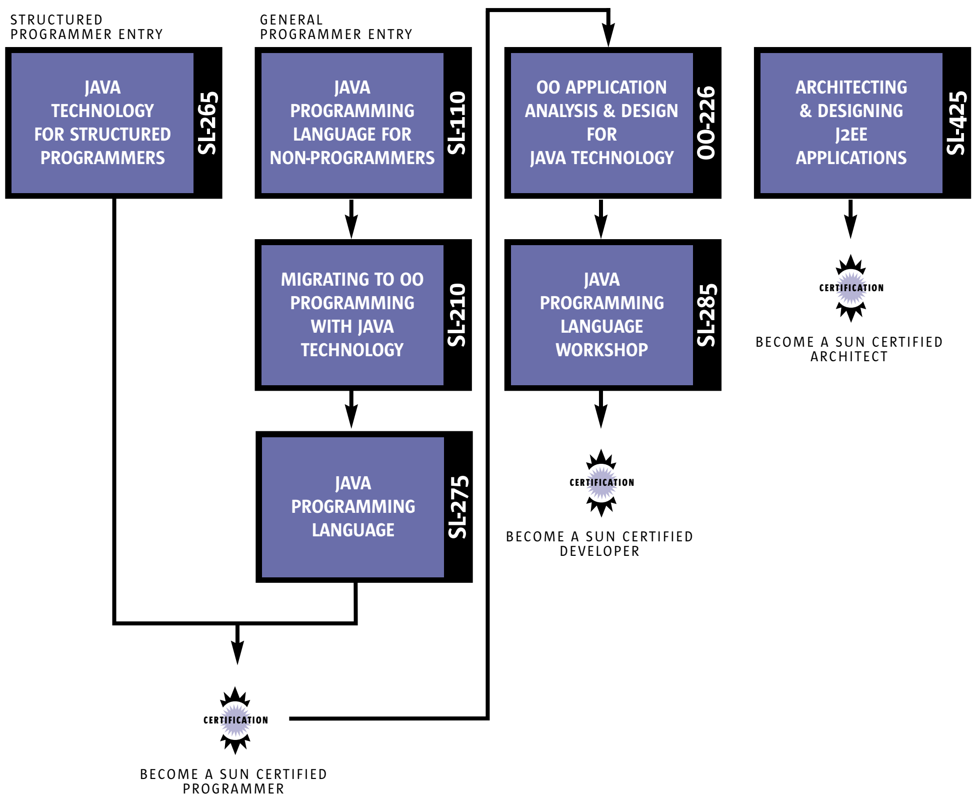
Java Technology for Structured Programmers
Course Number: SL-265
Duration: 5 days
Delivery: Instructor led

Java Programming Language for Non-Programmers
Course Number: SL-110
Duration: 5 days
Delivery: Instructor led

Migrating to OO Programming with Java Technology
Course Number: SL-210
Duration: 3 days
Delivery: Instructor led

Java Programming Language
Course Number: SL-275
Duration: 5 days
Delivery: Instructor led

JAVA TECHNOLOGY CERTIFICATION LEARNING PATH



Steps to Certification

1. PREPARE FOR TESTING

Review the exam objectives in this guide to help verify that you have sufficient knowledge to complete the exam successfully.

2. VERIFY THE EXAM NUMBERS AND REGISTER

When you're ready to register, you can purchase an exam voucher from your local Sun Educational Services office. To find your local office, go to http://www.sun.com/service/suned and choose the country in which you want to take the test.

3. CONTACT PROMETRIC TO SCHEDULE YOUR EXAM

The exams take place at authorized Prometric Testing Centers. To register for a convenient date, time, and location, go to http://www.2test.com for information on your local Prometric office. In some countries, you may register for exams online.

4. TAKE YOUR EXAM

Before starting the exam, you must agree to maintain test confidentiality and sign the Certification Candidate Pre-Test Agreement, which can be viewed at http://suned.sun.com/US/certification/register/policies.html. If you do not sign the agreement, you will not be allowed to take the exam.

After completing your exam, you will receive your score and a section-by-section assessment of your test performance. Your exam results will be available at http://suned.sun.com/US/certification/my_certification/index.html within three to five business days. If you do not pass the exam, you must wait two weeks before taking it again.

After you've successfully completed all of the certification requirements, you will receive a Welcome Kit from Sun, which includes a letter of congratulations, a personalized certificate, a lapel pin, and a logo license agreement, which, once signed, allows you to use the Java technology logo on your business card. At that time, you might want to start thinking about the next certification in your professional future. To begin preparation, visit http://suned.sun.com/US/certification/java/index.html to download the Certification Success Guide for the next certification in your career path.

Testing Objectives

Testing objectives for the Programmer for Java 2 Platform include:

DECLARATIONS AND ACCESS CONTROL

- Write code that declares, constructs, and initializes arrays of any base type using any of the permitted forms both for declaration and for initialization
- Declare classes, inner classes, methods, instance variables, static variables, and automatic (method local) variables making appropriate use of all permitted modifiers (such as public, final, static, abstract, and so forth)
- State the significance of each of these modifiers both singly and in combination, and state the effect of package relationships on declared items qualified by these modifiers
- For a given class, determine if a default constructor will be created, and if so, state the prototype of that constructor
- State the legal return types for any method given the declarations of all related methods in this or parent classes

FLOW CONTROL AND EXCEPTION HANDLING

- Write code using if and switch statements and identify legal argument types for these statements
- Write code using all forms of loops including labeled and unlabeled use of break and continue, and state the values taken by loop control variables during and after loop execution
- Write code that makes proper use of exceptions and exception handling clauses (try, catch, finally) and declares methods and overriding methods that throw exceptions

GARBAGE COLLECTION

- State the behavior that is guaranteed by the garbage collection system, and write code that explicitly makes objects eligible for collection

LANGUAGE FUNDAMENTALS

- Identify correctly constructed source files, package declarations, import statements, class declarations (of all forms including inner classes), interface declarations and implementations (for *java.lang.Runnable* or other interface described in the test), method declarations (including the main method that is used to start execution of a class), variable declarations and identifiers
- State the correspondence between index values in the argument array passed to a main method and command line arguments
- Identify all Java programming language keywords and correctly constructed identifiers
- State the effect of using a variable or array element of any kind when no explicit assignment has been made to it
- State the range of all primitive data types and declare literal values for String and all primitive types using all permitted formats, bases, and representations

OPERATORS AND ASSIGNMENTS

- Determine the result of applying any operator, including assignment operators, instance of, and casts to operands of any type, class, scope, or accessibility, or any combination of these
- Determine the result of applying the boolean equals (Object) method to objects of any combination of the classes *java.lang.String, java.lang.Boolean, and java.lang.Object*
- In an expression involving the operators &, |, &&, ||, and variables of known values, state which operands are evaluated and the value of the expression
- Determine the effect upon objects and primitive values of passing variables into methods and performing assignments or other modifying operations in that method

OVERLOADING, OVERRIDING, RUNTIME TYPE, AND OBJECT ORIENTATION

- State the benefits of encapsulation in object oriented design and write code that implements tightly encapsulated classes and the relationships "is a" and "has a"
- Write code to invoke overridden or overloaded methods and parental or overloaded constructors, and describe the effect of invoking these methods
- Write code to construct instances of any concrete class including normal top level classes, inner classes, static inner classes, and anonymous inner classes

THREADS

- Write code to define, instantiate, and start new threads using both *java.lang.Thread* and *java.lang.Runnable*
- Recognize conditions that might prevent a thread from executing
- Write code using synchronized, wait, notify, or notifyAll, to protect against concurrent access problems and to communicate between threads.
- Define the interaction between threads and between threads and object locks when executing synchronized, wait, notify, or notifyAll

THE JAVA.AWT PACKAGE

- Write code using component, container, and LayoutManager classes of the java.awt package to present a Graphical User Interface with specified appearance and resize behavior, and distinguish the responsibilities of layout managers from those of containers
- Write code to implement listener classes and methods, and in listener methods, extract information from the event to determine the affected component, mouse position, nature, and time of the event
- State the event classname for any specified event listener interface in the *java.awt.event* package

THE JAVA.LANG PACKAGE

- Write code using the following methods of the *java.lang.Math* class: abs, ceil, floor, max, min, random, round, sin, cos, tan, and sqrt
- Describe the significance of the immutability of string objects

THE JAVA.UTIL PACKAGE

- Make appropriate selection of collection classes/interfaces to suit specified behavior requirements

THE JAVA.IO PACKAGE

- Write code that uses objects of the file class to navigate a file system
- Write code that uses objects of the classes InputStreamReader and outputStreamWriter to translate between Unicode and either platform default or ISO 8859-1 character encoding and Distinguish between conditions under which platform default encoding conversion should be used and conditions under which a specific conversion should be used
- Select valid constructor arguments for FilterInputStream and FilterOutputStream subclasses from a list of classes in the *java.io.package*
- Write appropriate code to read, write and update files using FileInputStream, FileOutputStream, and RandomAccessFile objects
- Describe the permanent effects on the file system of constructing and using FileInputStream, FileOutputStream, and RandomAccessFile objects

Sample Questions

Please visit http://suned.sun.com/US/certification/java/java_progj2se.html and click on ePractice Sample Questions for the Sun Certified Programmer for Java 2 Platform.